Hybrid Mesh Micro-Climate Wireless Sensor Network

Comprehensive Technical Overview

Design Practicum Project
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All engineering claims cross-checked against datasheets, RFC standards, and independent technical research.

Key Technologies

• LoRa 865 MHz (IN865)	• TinyML / XGBoost	• AES-128-CCM
• AODV Mesh Routing	• Spatial Kriging	• LiFePO4 Power System
• ESP32-S3 WROOM	• SX1262 Transceiver	• React/Leaflet PWA

Slave Node Est. Cost: Rs. 12,060 | Master Node Est. Cost: Rs. 14,210

About This Project

This project is a ground-up engineering effort to design, build, and deploy a **self-sustaining, solar-powered, AI-augmented Wireless Sensor Network (WSN)** capable of performing continuous, high-resolution micro-climate monitoring across a macro-regional geographic area — without any reliance on cellular infrastructure, cloud services, or paid spectrum.

The core innovation is the fusion of three independent technologies into a single coherent system:

- **Long-Range RF (LoRa):** Sub-GHz radio reaching 5–15 km in open terrain on milliwatts of power, using the free 865–867 MHz ISM band allocated by India's WPC Wing.
- **Embedded Machine Learning (TinyML):** A pre-trained XGBoost decision-tree model compiled into each sensor node's firmware, enabling real-time local anomaly detection without cloud dependency.
- **Reactive Mesh Networking (AODV):** A lightweight, on-demand routing protocol forming a self-healing mesh that survives node failures and RF obstructions without a centralised access point.

Each sensor node (**Slave Node**) is a weatherproofed, IP67-sealed embedded system built around the ESP32-S3 microcontroller, hosting an array of 11 environmental sensors. It runs on a 10 W solar panel and a LiFePO4 battery with an average current draw verified at under 0.5 mA.

A single **Master Node** aggregates data from all Slave Nodes, performs Spatial Kriging interpolation, and serves results via a lightweight JSON REST API over a local WiFi Access Point. The administrator accesses a React/Leaflet Progressive Web App (PWA) to view live weather heatmaps and inject RPC commands into any mesh node.

Primary application domains: severe weather early warning, wildfire smoke detection, illegal logging detection (acoustic AI), agricultural drought mapping, and ambient radiation monitoring.

1. Executive Summary

This project designs and validates a **Hybrid Mesh WSN** for macro-regional micro-climate mapping and severe weather anomaly detection. By coupling long-range LoRa radio with on-device TinyML inference (XGBoost), the network eliminates cloud dependency entirely. Each node operates autonomously on solar power, runs local anomaly detection, and remains radio-silent during normal conditions. Only genuine weather anomalies trigger unsolicited emergency transmissions.

1.1 Advantages Over Conventional WSN Systems

Table 1: System Comparison

Criterion	Conventional Cellular WSN	This System
Recurring Cost	Rs. 200–500/node/month (SIM)	Rs. 0 (ISM band, license-exempt)
Infrastructure Dep.	Dies when cell towers fail	Fully P2P; mesh survives independently
Spatial Resolution	km-scale between stations	Meter-scale; deployable anywhere

Criterion	Conventional Cellular WSN	This System
Battery Life	6–18 months (coin/AA cell)	Indefinite (10 W solar + LiFePO4)
Intelligence	Dumb data relay; cloud processing	On-device XGBoost anomaly gating
Failure Tolerance	Single gateway = SPOF	AODV self-heals around dead nodes

2. System Architecture & Topology

The network deliberately abandons the LoRaWAN star topology in favour of a **Hybrid Mesh**. In a star topology, all nodes must be within direct radio range of a central gateway. In this mesh, Slave Nodes relay packets for each other using AODV routing, so nodes several kilometres beyond the Master's direct reach remain fully connected.

2.1 Architectural Zones

- **Zone 1 — Admin Client Device:** Runs an offline-first React/Leaflet PWA. Fetches lightweight JSON from the Master's REST API. No server-side code executes on this device.
- **Zone 2 — Master Node (ESP32-S3-N16R8):** Maintains dual radio interfaces — WiFi Access Point (for admin PWA) and SX1262 LoRa transceiver (for the mesh). Its Spatial Kriging engine interpolates weather data between physical sensor locations. A DS3231 hardware RTC maintains UTC timestamps that synchronize network-wide polling windows.
- **Zone 3 — Slave Mesh:** Field-deployed array of identical sensor nodes. Communicate exclusively over LoRa RF. WiFi and Bluetooth are disabled during normal operation to preserve power.

3. Slave Node Firmware — Sequential Execution Architecture

3.1 Design Philosophy

The slave node firmware is implemented as a **Monolithic Sequential Task Flow** within a single FreeRTOS task. This is a deliberate engineering choice to honour the strict power budget. Using multiple FreeRTOS tasks communicating via queues introduces:

- **Task Stack RAM overhead:** Each task requires its own stack (2–8 KB), consuming precious SRAM.
- **Context-switch CPU cycles:** The RTOS scheduler prevents full clock gating, extending active time.
- **Extended active window:** CPU remains powered longer for inter-task coordination, destroying the sleep-current budget.

A sequential pipeline executes fully and deterministically before calling `esp_deep_sleep_start()`. The entire active cycle completes in under 200 ms.

3.2 Step-by-Step Execution

Step 1 — Wake: RTC 5-minute timer or GPIO interrupt (AS3935 / Pluviometer) exits deep sleep.

Step 2 — Boot Init: Load AES-128-CCM pre-shared key and rolling nonce from RTC RAM (persists through deep sleep).

Step 3 — Power Sensors: Assert MOSFET gate pins; wait 50 ms for I2C bus stabilisation.

Step 4 — Read Sensors: Sequential I2C/UART/ADC reads across 11 sensors (~80 ms total). Packed into SensorDataStruct.

Step 5 — Preprocess: Min-Max scaling into 22-float 1D tensor (11 absolute + 11 delta values using RTC RAM stored previous readings).

Step 6 — TinyML Inference: XGBoost decision tree on the 1D tensor; outputs 8-bit Anomaly Index (0=Normal, 1=Rain, 2=Thunderstorm, 3=High PM2.5, 4=Radiation Spike). Runtime ~15 ms at 240 MHz.

Step 7 — AI Gate: If Index > 0: proceed to Crypto TX (unsolicited anomaly alert). If Index = 0: check RTC for synchronized polling window (2 s window at top of every hour). If in window, open LoRa RX, await CMD_SEND_TELEMETRY. If polled → Crypto TX. Otherwise → Sleep immediately.

Step 8 — Crypto TX: Encrypt and authenticate payload with AES-128-CCM; atomically increment nonce in RTC RAM.

Step 9 — LoRa TX + ACK: SPI DMA transfer to SX1262; await link-layer ACK (~500 ms timeout). Packet ToA ~103–118 ms at SF8/BW125 for 30-byte payload.

Step 10 — Deep Sleep: esp_deep_sleep_start(). ULP resets 5-minute timer. Total draw collapses to ~63 µA (ESP32-S3 ULP ~8 µA + XC6220 LDO Iq ~55 µA).

4. Power System Design & Autonomy Validation

4.1 Battery Chemistry — LiFePO4 (Mandatory)

Critical Warning — Battery Safety

Standard NMC/NCA Li-Ion 18650 cells are **explicitly prohibited**. Indian summer ambient temperatures reach 45–48°C. Inside a sealed sun-facing IP67 ABS enclosure, internal temperatures exceed **65–75°C**. NMC Li-Ion degrades above 45°C and enters thermal runaway above ~70°C. NMC offers only 300–500 charge cycles vs. 2,000–3,000 for LiFePO4.

- Thermally stable to ~270°C (vs. ~150°C for NMC)
- 2,000–3,000 charge cycles with <20% capacity loss
- Nominal 3.2 V / charge termination 3.65 V — requires LiFePO4-specific MPPT controller (CN3722)

4.2 MPPT Charge Controller — CN3722

The CN3722 (CONSONANCE Semiconductor) supports LiFePO4 charging via external resistor divider on the FB pin, configuring charge termination to exactly 3.65 V/cell. Charge current is set via RPROG resistor (configured for 1 A in this design).

Engineering Note

The 10 W/12 V monocrystalline panel's open-circuit voltage (Voc) is ~21 V; Vmpp ≈ 17.5 V. The CN3722's maximum input is 6 V. An **LM2596 step-down pre-regulator** (set to 5.5 V output) is required between the panel and CN3722. This component is explicitly included in the BOM.

4.3 Voltage Regulation — XC6220B331MR (1 A LDO)

The XC6220B331MR (Torex Semiconductor) was selected over the HT7333 because the SX1262 LoRa radio draws up to **118 mA at +22 dBm TX** (verified from Semtech SX1262 datasheet Rev 2.1). Peak combined draw during TX is 200–250 mA. The XC6220's 1 A rating provides >4× headroom. The HT7333's 250 mA limit causes guaranteed brownout resets on every LoRa transmission.

4.4 Power Autonomy Calculation

Table 2: Average Current Budget per 5-Minute Operating Cycle

State	Current Draw	Duration / 5-min Cycle	Avg Contribution
Deep Sleep (ULP + LDO)	~63 µA	299.8 s	0.063 mA
Sensor Read (CPU + I2C)	~120 mA	0.08 s	0.032 mA
XGBoost Inference	~80 mA	0.015 s	0.004 mA
LoRa TX (SF8, 22 dBm)	~120 mA	0.118 s	0.047 mA
LoRa RX (Poll Window)	~12 mA	2 s (1×/hour)	0.001 mA
TOTAL AVERAGE	—	—	~0.147 mA

- **Battery:** 32650 LiFePO4, 6,000 mAh at 3.2 V nominal
- **Autonomy (no solar):** 6,000 ÷ 0.147 ≈ 40,816 h ≈ 4.7 years
- **Solar input (clear sky):** 10 W × 5 h/day = 50 Wh/day; daily node use ≈ 0.011 Wh — panel output is >4,500× daily consumption
- **Solar input (heavy monsoon):** Realistically 5–10 Wh/day — still >450× daily node consumption

5. Peer-to-Peer Mesh Routing — AODV

5.1 Why Not Distance-Vector (DV)?

Classic DV routing requires every node to proactively broadcast its entire routing table at regular intervals. In a 50-node network, each broadcast is ≥50 bytes transmitted continuously by every node — creating a **broadcast storm** that destroys LoRa channel capacity. DV is architecturally incompatible with low-bandwidth LoRa.

5.2 AODV — Ad-hoc On-Demand Distance Vector (IETF RFC 3561)

AODV is a *reactive* protocol. Routes are only discovered when needed — no periodic routing table broadcasts. The discovery sequence:

- **Route Discovery:** Node A broadcasts a RREQ (Route Request) only when it needs a route.
- **Propagation:** Relay nodes forward the RREQ until it reaches the destination (Master).
- **Confirmation:** The Master sends a RREP (Route Reply) back on the exact discovered path.
- **Execution:** Node A caches the route (3,600 s lifetime) and sends its payload.
- **Self-Healing:** Failed links trigger RERR packets; nodes invalidate stale routes and re-discover automatically within ~500 ms.

5.3 Custom LoRa-Adapted AODV Packet Sizes

Standard RFC 3561 uses 4-byte IPv4 addresses (RREQ=24 B, RREP=20 B). For this LoRa implementation, IPv4 addresses are replaced with 2-byte node IDs, and a 2-byte software CRC-16 checksum is appended to all packets (AODV control and data):

Table 3: Custom Binary AODV Packet Sizes

Packet	Fields	Total
RREQ	Type(1)+Flags(1)+HopCount(1)+BcastID(2)+DestID(2)+DestSeq(2)+OrigID(2)+OrigSeq(2)+Checksum(2)	15 bytes
RREP	Type(1)+Flags(1)+HopCount(1)+DestID(2)+DestSeq(2)+OrigID(2)+Lifetime(2)+Checksum(2)	13 bytes
RERR	Type(1)+DestID(2)+ErrorCode(1)+Checksum(2)	6 bytes

At SF8/BW125, a 15-byte RREQ has a ToA of approximately 77 ms — negligible airtime for an on-demand operation. Intermediate relay nodes verify this software CRC-16 (using the standard CRC-16-CCITT polynomial 0x1021) first; if it fails, the packet is immediately dropped, conserving CPU resources.

6. Security & Cryptography

6.1 AES-128-CCM (Counter with CBC-MAC)

AES-128-CCM is standardised in NIST SP 800-38C and IETF RFC 3610. It is the cryptographic foundation of IEEE 802.15.4 (Zigbee) and LoRaWAN 1.1 payload encryption. It achieves both confidentiality and authenticity in a single pass:

- **Confidentiality:** Payload encrypted using AES in Counter (CTR) mode.
- **Authenticity:** 8-byte Message Integrity Code (MIC) generated via CBC-MAC. RFC 3610 permits tag lengths of 4, 6, 8, 10, 12, 14, or 16 bytes. The 8-byte tag offers 2■■ possible values — forgery is computationally infeasible.

Engineering Note

LoRaWAN comparison: Standard LoRaWAN 1.1 uses AES-CMAC with a 4-byte MIC. This project is a **custom private mesh network**, deliberately selecting an 8-byte MIC for stronger authentication. Total packet overhead: 8 bytes (AES-128-CCM MIC) + 2 bytes (software CRC-16 checksum) = 10 bytes. HMAC-SHA256 would add 32 bytes — a 300% overhead increase unacceptable on a LoRa link.

6.2 Nonce Management

- 32-bit nonce stored in **RTC RAM** (persists through deep sleep).
- Atomically incremented after each encrypted transmission.
- Backed up to NVS flash every 100 transmissions.
- New node integration uses a **challenge-response handshake** under a factory-provisioned bootstrap key to assign a fresh nonce range without replay exposure.

6.3 Secure RPC Execution

All RPC commands are authenticated. The Slave: (1) decrypts with AES-128-CCM, (2) validates the 8-byte MIC — a single-bit mismatch causes silent rejection, (3) verifies nonce is within expected window (±5), (4) only then executes the command.

7. Master Node — Spatial Kriging Engine

7.1 What is Spatial Kriging?

Ordinary Kriging (Matheron, 1963) is a Gaussian process regression technique that estimates values at unobserved spatial locations based on surrounding measurements:

$z^*(x^*, y^*) = \Sigma\, \lambda\blacksquare \cdot z\blacksquare$ (i = 1..N)

Weights $\lambda\blacksquare$ are computed by solving an (N+1)×(N+1) system of linear equations (minimising estimation variance) using a spatial variogram model.

7.2 Feasibility on ESP32-S3

- For N = 10–20 nodes: matrix is 21×21 = 441 floats = 1.7 KB RAM
- Gaussian elimination: $O(N^3) \approx$ 9,261 multiply-accumulate operations — completes in <5 ms at 240 MHz
- 8 MB PSRAM (ESP32-S3-N16R8) provides sufficient working memory. Active computation should use internal SRAM (~512 KB); PSRAM is ~10× slower (SPI-accessed).
- Pre-compute the covariance matrix A (based on fixed node positions) once at startup. Only solve $\Lambda = A\blacksquare^{1}b$ at runtime — reduces per-reading compute to <1 ms.

7.3 Predictive Radiation Level & Plume Dispersion Module

To provide early forecasting rather than purely reactive warnings, the network integrates a two-stage **Predictive Radiation Level Modeling** module split between edge nodes and the Master.

- **Poisson Noise Filtering:** The BPW34 PIN diode has a small active surface area (7.5 mm^2), resulting in low background counts per minute (CPM) and high statistical Poisson noise. The Slave Node runs an embedded **One-Dimensional Kalman Filter** to smooth out statistical fluctuations and estimate the true local CPM background level without lag.
- **Radon Washout Prediction:** During heavy precipitation, natural radon decay products are washed out from the atmosphere, temporarily spiking the local radiation background by up to 2× (Radon Washout). To prevent false alarms, the node's XGBoost model uses inputs from the Pluviometer and the BME680 barometric pressure to predict expected precipitation-induced radiation spikes: $CPM\_expected = \alpha \times RainRate + \beta \times BarometricPressure + Baseline$. If the measured CPM exceeds CPM_expected by a pre-set threshold (e.g., 3σ), the node classifies it as a true nuclear/industrial anomaly and initiates an emergency alert.
- **Master Node Dispersion Prediction (Gaussian Plume Model):** When a Slave Node reports a true, verified radiation anomaly, the Master Node executes a **Gaussian Plume Dispersion Model** to predict the future geographic spread of the radioactive plume across the region. It pulls the wind speed (u) from the reporting node's Anemometer and the absolute wind direction (θ) from the AS5600 wind vane to calculate the predicted ground-level concentration C(x,y) downwind over a 1–6 hour horizon: $C(x,y) = Q / (\pi\, u\, \sigma\_y\, \sigma\_z) * \exp(-y^2/2\sigma\_y^2 - H^2/2\sigma\_z^2)$. This predicted dispersion grid is fed into the Spatial Kriging engine and served as a GeoJSON contour layer to the Leaflet PWA, allowing emergency services to visualize the predicted fallout zone in real-time.

8. Sensor Array — Full Technical Specifications

Table 4: 11-Sensor Micro-Climate Array (Identical on Master and Slave Nodes)

#	Sensor	Protocol	Parameters	Engineering Notes
1	BME680	I2C (0x76)	Temp (±1°C), RH (±3%), Pressure (±1 hPa), VOC IAQ	25 ms measurement cycle; integrated heater for gas sensing.
2	AS3935	I2C/SPI	Lightning distance 1–40 km, energy estimate	EMI Critical: grounded copper Faraday shield MANDATORY inside PCB ground pour. SX1262 at 22 dBm causes false-positive interrupts without shielding.
3	NEO-6M GPS	UART 9600	Latitude, Longitude, UTC (±1 µs)	Used once at deployment for geo-registration. UTC thereafter from DS3231. Power-gated after use.
4	Anemometer	GPIO/PCNT	Wind speed (m/s) via pulse frequency	A3144 Hall-Effect sensor. Each magnet pass = 1 PCNT interrupt. Speed = frequency × calibration constant.
5	Pluviometer	GPIO IRQ	Precipitation (mm) via bucket tips	Reed switch; 1 tip = 0.2 mm. Stays powered at nano-amp level during deep sleep for emergency GPIO wake.
6	BPW34 + LM358	ADC	Beta/Gamma radiation (count rate)	~1 nA ionisation pulse amplified by LM358 TIA. Validated in IEEE embedded dosimeter literature. Must be optically shielded (light-tight).
7	INMP441 + LM393	I2S + GPIO	Acoustic classification (thunder, chainsaw)	LM393 comparator (µA quiescent) triggers GPIO wake on threshold. INMP441 powers on for 500 ms FFT window only.
8	PMS5003	UART 9600	PM1.0, PM2.5, PM10 (µg/m³)	Internal fan ~80 mA. Power-gated. 500 ms fan warm-up required before valid readings.
9	AS5600	I2C (0x36)	Wind vane direction 0–359°, 12-bit	Contactless magnetic encoder; fixed I2C address 0x36 (non-configurable). Zero friction wear.
10	LTR390-UV	I2C (0x53)	UVA, Ambient Light (Lux)	Digital UV Sensor: Measures UVA light and ambient light. Onboard ADC and logic outputs counts to compute UV index and lux. Replacing obsolete VEML6075.
11	SHT20	I2C (0x40)	Soil temp (±0.3°C), Soil moisture (%RH)	CLARIFICATION: SHT20 is natively I2C. RS-485 is a module-level wrapper in industrial transmitters. Direct I2C wiring via potted waterproof probe used here.

9. Master-Slave RPC Command Set

All commands are transmitted as encrypted AES-128-CCM payloads. The Slave verifies MIC and nonce before execution.

Listing 1: RPC Opcode Enumeration (firmware/rpc.h)

```
enum RpcCommand : uint8_t {
    CMD_FORCE_INFERENCE  = 0x01, // Force immediate sensor read + XGBoost
    CMD_SEND_TELEMETRY   = 0x02, // Reply with full SensorDataStruct (poll)
    CMD_UPDATE_THRESH    = 0x03, // Update XGBoost anomaly threshold
    CMD_SET_SLEEP_PERIOD = 0x04, // Change deep sleep interval (seconds)
    CMD_DISABLE_PERIPH   = 0x05, // Cut MOSFET power to broken sensor
    CMD_REBOOT           = 0x06, // Issue esp_restart() for OTA recovery
    CMD_SYNC_NONCE       = 0x07, // Resync nonce after power-loss reset
    CMD_GET_BATTERY      = 0x08  // Report battery voltage via ADC
};
```

10. Engineering Design Standards

10.1 Cabling

Table 5: Wire Gauge and Insulation Specifications

Line Type	Gauge	Insulation	Rationale
Power (Solar → MPPT → Battery)	22 AWG	Silicone	Handles up to 5 A; resists 200°C; flexible at –40°C
Signal (I2C, SPI, UART, GPIO)	26 AWG	Silicone	Capacitance <30 pF/m; prevents I2C ACK corruption at 400 kHz
Antenna (SMA bulkhead)	RG-174 / LMR-100	PTFE coax	Maintains 50 Ω impedance; PTFE rated to 260°C

10.2 Connectors

Table 6: Connector Specifications

Application	Connector	Rationale
Sensor Board-to-Wire	JST-XH 2.54 mm latched	Latching prevents vibration disconnects; polarised against reverse connection
Battery Power Rail	XT60H panel-mount	Rated 60 A; gold-plated; polarised housing prevents shorts
RF Antenna	IPEX/U.FL to SMA-F bulkhead	SMA-F through enclosure wall; sealed with neoprene O-ring

10.3 Mechanical & Enclosure

Table 7: Mechanical Specifications

Item	Specification	Rationale
Enclosure	IP67-rated ABS, min 120×80×55 mm	IP67 = 30 min immersion to 1 m depth

Item	Specification	Rationale
Screws	M3 SS304 Socket Cap	SS304 corrosion-immune; zinc-plated screws fail in 6–18 months outdoors
Threaded Inserts	M3 Brass heat-set (PETG/ASA mounts)	PETG/ASA self-tapping threads strip after 3–5 uses
PCB Standoffs	M3 Nylon Hex (10 mm)	Electrically isolates PCB from enclosure floor
Acoustic Vents	IP67-rated PTFE membrane	Pressure equalisation without water ingress
Sealant	Neutral-cure silicone ONLY	Acetoxy-cure releases acetic acid during cure, corroding PCB copper traces

11. Bill of Materials (BOM)

Critical Warning

Battery chemistry: Standard Li-Ion/NMC 18650 cells are prohibited. All cells must be LiFePO4. **LTR390-UV Note:** Digital I2C sensor measuring UVA light and ambient light, replacing the obsolete VEML6075 for stable long-term sourcing. **LM2596 pre-regulator:** Required between solar panel (Voc ~21 V) and CN3722 (max 6 V input). Included in BOM below.

11.1 Slave Node BOM

Table 8: Slave Node Bill of Materials (Single Unit, Prototype Pricing)

Category	Component	Qty	Price (Rs)
Compute	ESP32-S3 WROOM-1 (N8, 8 MB flash)	1	650
RF	SX1262 SPI Module (865–867 MHz)	1	750
	3 dBi Omni Dipole Antenna (SMA)	1	200
	IPEX/U.FL to SMA-F Bulkhead Pigtail (RG-174)	1	150
Power	10 W Monocrystalline Solar Panel	1	900
	LM2596 Buck Pre-regulator (5.5 V out)	1	80
	CN3722 MPPT Charge Controller (LiFePO4)	1	450
	32650 LiFePO4 Cell (3.2 V, 6,000 mAh)	1	600
	XC6220B331MR LDO (1 A, 3.3 V, Iq=55 µA)	1	60
	IRLML2502 N-Ch MOSFET (Vgs(th) <2 V) ×4	4	100
	Decoupling Caps (100 µF + 100 nF per rail)	—	50
Sensors	BME680 Module (I2C 0x76)	1	950
	AS3935 Lightning IC + MA5532 Antenna Coil	1	1,840
	NEO-6M GPS Module (UART)	1	450

Category	Component	Qty	Price (Rs)
	BPW34 PIN Photodiode	1	100
	LM358 Dual Op-Amp (DIP-8) — Radiation TIA	1	50
	PMS5003 Particulate Sensor (UART)	1	1,200
	INMP441 I2S MEMS Microphone	1	150
	LM393 Dual Comparator (DIP-8) — Acoustic	1	30
	AS5600 Magnetic Encoder Module	1	250
	LTR390-UV Digital UV Sensor	1	750
	SHT20 Waterproof Probe (I2C, potted cable)	1	400
	A3144 Hall-Effect Sensor	1	30
	Reed Switch SPST-NO ×2	2	50
Mechanical	IP67 ABS Enclosure (120×80×55 mm)	1	600
	M3 Brass Heat-Set Inserts ×8	8	50
	M3 SS304 Screws + M2.5 Nylon Standoffs	—	90
	IP67 PTFE Acoustic/Pressure Vents ×2	2	200
	Copper Tape (AS3935 Faraday Shield)	1	100
	Neutral-Cure Silicone Sealant	1	80
	Silica Gel Desiccant 5 g ×2	2	20
	22 AWG Silicone Wire (5 m)	1	100
	26 AWG Silicone Wire (5 m)	1	80
	JST-XH 2.54 mm Connector Kit	1	100
	XT60H Connector Pair	1	80
	ASA/PETG Filament (0.1 kg)	0.1 kg	120
TOTAL			Rs. 12,060

11.2 Master Node Additional Components

Table 9: Master Node Additional Components

Category	Component	Qty	Price (Rs)
Compute Upgrade	ESP32-S3-N16R8 (16 MB Flash / 8 MB PSRAM)	1	700
RF Upgrade	8 dBi Fiberglass Omni Antenna (868 MHz, SMA)	1	800
Storage	MicroSD SPI Module + 32 GB MicroSD (Class 10)	1	500
Timing	DS3231 Precision RTC Module (I2C + CR2032)	1	150
Additional Total			Rs. 2,150

Category	Component	Qty	Price (Rs)
Master Node Total	Slave BOM + Additions		Rs. 14,210

Critical Warning

DS3231 Counterfeit Risk: Authentic DS3231 achieves ±2 ppm accuracy (TCXO). Counterfeit modules on AliExpress/Amazon lack the genuine TCXO and drift 50–100× faster (seconds per day instead of milliseconds). **Verify chip authenticity before deployment.**

12. Bandwidth & RF Physics Validation

12.1 LoRa Configuration & Time-on-Air

Selected LoRa Parameters (IN865 band plan, LoRa Alliance Regional Parameters v1.0.3): Frequency 865.0625 MHz (IN865 Channel 0), SF8 default (SF9 for longer relay hops), BW 125 kHz, CR 4/5, Max Payload 222 bytes at SF8.

Table 10: Time-on-Air (Independently Verified via Semtech LoRa Calculator)

Spreading Factor	Payload	Bandwidth	Coding Rate	Time-on-Air
SF7	25 bytes	125 kHz	4/5	~46–52 ms
SF7	30 bytes	125 kHz	4/5	~52–58 ms
SF8	25 bytes	125 kHz	4/5	~92–103 ms
SF8	30 bytes	125 kHz	4/5	~103–118 ms (design value)

SF8 is the default — 2.5 dB more link budget than SF7 for improved resilience, at ~2× airtime. Self-imposed duty cycle: 118 ms / 300,000 ms = **0.039%** — well below Europe's 1% rule.

12.2 Indian Regulatory Compliance

Table 11: WPC/DoT Regulatory Compliance

Parameter	WPC/DoT Requirement	This System
Frequency Band	865–867 MHz (license-exempt SRD)	865.0625 MHz — Compliant
Maximum EIRP	1 W (30 dBm)	22 dBm (158 mW) — Compliant
Duty Cycle	No mandatory limit	Self-imposed 0.039% — Compliant
Licence Requirement	None	None — Compliant

12.3 Range Estimation

Table 12: LoRa Range Estimates (865 MHz, Friis + COST-231 Hata Model)

Environment	SF7 Range	SF8 Range
Open terrain (line-of-sight)	2–5 km	5–10 km
Sub-urban (partial obstruction)	1–3 km	2–4 km

Environment	SF7 Range	SF8 Range
Dense forest / urban canyon	0.3–1 km	0.5–1.5 km

12.4 Solar Power — Monsoon Realism

Table 13: Realistic 10 W Solar Panel Daily Output in India

Sky Condition	Daily Output (10 W Panel)
Clear sky (summer)	40–50 Wh/day
Light overcast	20–35 Wh/day
Heavy monsoon cloud/rain	5–10 Wh/day
Extended storm (3+ days)	Near-zero

Node daily consumption: ~0.011 Wh. Even at 5 Wh/day monsoon output, the panel produces >450× the node's daily energy requirement. The 6,000 mAh LiFePO4 battery provides 4.7 years backup with zero solar input.

13. Web Interface Architecture

The Master Node is a **stateless JSON REST API only**. It does not serve HTML/CSS/JS bundles. The ESP32-S3's lwIP TCP/IP stack is optimised for small, frequent API requests — not for serving multi-megabyte static assets.

13.1 API Endpoints

Table 14: REST API Endpoint Design

Endpoint	Method	Description
/api/nodes	GET	All registered node IDs and last-seen timestamps
/api/nodes/{id}/telemetry	GET	Latest sensor readings from node {id}
/api/kriging/heatmap	GET	Kriging-interpolated weather grid as GeoJSON
/api/rpc/{id}	POST	Inject an RPC command to node {id} in the mesh
/api/status	GET	Master health: battery voltage, uptime, mesh node count

13.2 Client PWA

- React 18 + Leaflet.js Progressive Web App; pre-built static bundle installed on admin's device via Service Worker
- Leaflet.heat plugin renders Kriging GeoJSON as colour gradient heatmap overlay on the base map
- Last-fetched data stored in IndexedDB for offline map review when not connected to Master WiFi AP
- RPC injection via POST /api/rpc/{id} — all 8 commands triggerable from node marker popup in the UI

14. Development Work Pipeline

Table 15: Phased Development Pipeline (6 Phases, 40 Tasks)

Phase	#	Task	Deliverable	Acceptance Criteria
0 — Procure	0.1	Order all BOM components (Slave ×4, Master ×1)	Components on bench	All verified against BOM; DOA identified
	0.2	Validate LiFePO4 + CN3722 charge curve	Oscilloscope profile at 1 A	Terminates at 3.65 V ±50 mV
	0.3	Verify XC6220 under 300 mA step-load	Bench PSU + multimeter log	3.3 V ±50 mV; no drop below 3.0 V
	0.4	Verify LM2596 output (21 V in, 5.5 V out)	Multimeter	Stable 5.5 V ±0.1 V at 500 mA
1 — PCB	1.1	Slave PCB schematic (KiCad)	.sch file	ERC passes; zero errors
	1.2	Slave PCB layout (2-layer, ENIG, Cu AS3935 shield)	.kicad_pcb	DRC passes; Faraday pour around AS3935
	1.3	Master PCB (slave + DS3231 + MicroSD)	.kicad_pcb	DRC passes; SD SPI traces length-matched
	1.4	Fabricate PCBs (4× slave, 1× master)	Fabricated bare PCBs	No bridging; correct stack; ENIG finish
	1.5	Assemble slave nodes (4×)	4× assembled boards	No cold joints; visual inspection pass
	1.6	Assemble master node (1×)	1× assembled board	Same + DS3231 and MicroSD slot verified
	1.7	3D-print anemometer, vane, brackets (PETG/ASA)	Printed parts	Fit verified; no warping; UV-coat applied
2 — Firmware	2.1	I2C HAL: BME680, AS5600, LTR390-UV, SHT20	hal_i2c.cpp	Valid readings on serial monitor
	2.2	UART HAL: NEO-6M, PMS5003	hal_uart.cpp	GPS fix <60 s; PM2.5 valid after 500 ms
	2.3	GPIO/PCNT: Anemometer, Pluviometer, AS3935, LM393	hal_gpio.cpp	Pulse counter correct; interrupt on bucket tip
	2.4	ADC HAL: BPW34/LM358 radiation	hal_adc.cpp	Stable baseline; spike detectable
	2.5	MOSFET power gating + stabilisation delays	power_gate.cpp	No I2C bus lockup after gate cycle
	2.6	Min-Max preprocessing pipeline	preprocess.cpp	Tensor in [0.0, 1.0]; deltas correct
	2.7	XGBoost training data collection (2+ weeks)	training_data.csv	≥500 labelled samples; 5 anomaly classes
	2.8	Train, validate, tune XGBoost model	model_final.ubj	F1 ≥0.88; confusion matrix approved

Phase	#	Task	Deliverable	Acceptance Criteria
	2.9	Port XGBoost to C++ embedded inference	tinyml.cpp	Output matches Python reference on same inputs
	2.10	AES-128-CCM + nonce manager	crypto.cpp	Roundtrip pass; nonce survives sleep; NVS backup
	2.11	AODV routing (RREQ/RREP/RERR, route cache)	aodv.cpp	Route in ≤3 hops; RERR triggers re-discovery
	2.12	Synchronized polling window (RTC + LoRa RX)	poll_mgr.cpp	RX only within 2 s window; responds to CMD_SEND_TELEMETRY
	2.13	Full sequential pipeline + deep sleep	main.cpp	Cycle <200 ms; sleep <100 µA
	2.14	Challenge-response new-node handshake	handshake.cpp	New node joins; nonce sync; no replay
3 — Master	3.1	Spatial Kriging engine (variogram + Gauss elim.)	kriging.cpp	Output matches Python pykrige on 10-node grid
	3.2	Master polling scheduler (DS3231 + RPC broadcast)	scheduler.cpp	All nodes polled; responses logged to SD
	3.3	JSON REST API (5 endpoints, lwIP HTTP)	api.cpp	All endpoints correct; 10 req/s stress test
	3.4	Node registry + MicroSD database writer	registry.cpp	Telemetry persisted; recoverable after reboot
4 — Web PWA	4.1	React + Leaflet PWA frontend	build/ dist	Heatmap renders from live GeoJSON
	4.2	RPC injection UI (node click → POST)	RPC panel	All 8 commands injectable; response shown
	4.3	Service Worker offline packaging	sw.js + manifest	Installs on Chrome/Safari; offline data viewable
5 — Integrate	5.1	Bench test: 4 slaves + 1 master	Integration log	All nodes in /api/nodes; all readings valid
	5.2	AODV multi-hop relay test (D→B→A→Master)	Packet trace	3-hop delivery; re-routes on node kill
	5.3	Power budget validation (µA logger)	Current log	Sleep <100 µA; active <200 ms
	5.4	RF range test (RSSI at 1 km and 2 km)	RSSI/distance map	Link OK at 2 km SF8; SF9 at 4 km
	5.5	Security: MIC rejection + nonce replay test	Test report	Corrupted MIC rejected; replay rejected
6 — Deploy	6.1	Weatherproof assembly; seal enclosures	5× field units	IP67 submersion test (30 min, 0.5 m) passed
	6.2	Mount anemometers and vanes; calibrate	Assembled stations	Full 0–360° AS5600 reading; cups spin

Phase	#	Task	Deliverable	Acceptance Criteria
	6.3	Deploy and geo-register nodes	GPS coordinates logged	NEO-6M fix <60 s; in NVRAM; on PWA map
	6.4	72-hour field validation run	Field log + heatmap	Zero crashes; polling works; kriging generates
	6.5	Final documentation + report	This document	All decisions documented; in version control

15. Known Limitations & Open Engineering Items

- **BPW34 Radiation Calibration.** The BPW34 silicon PIN photodiode is an uncalibrated relative counter. While the predictive model filters statistical noise and compensates for radon washout, it cannot serve as a certified dosimeter without individual chamber calibration against a known reference source (e.g., Cesium-137).
- **LM2596 Pre-regulator.** Required between solar panel Vmpp (~17.5 V) and CN3722 (max 6 V input). Now explicitly included in BOM.
- **SHT20 Interface Clarification.** The SHT20 chip is natively I2C. RS-485 is a module-level wrapper in industrial transmitters. Direct I2C wiring is used here.
- **AS3935 False-Positive Calibration.** Even with Faraday shielding, the AFE_GB gain register and indoor/outdoor mode bit must be calibrated per deployment environment.
- **AODV on LoRa — No Reference Implementation.** A production-quality AODV stack for ESP32+LoRa must be built from scratch. This is the highest firmware complexity item in the project.
- **XGBoost Cold-Start Problem.** The model requires 2+ weeks of labelled training data. Until data exists, use a conservative rule-based threshold as a placeholder (e.g., temperature delta >5°C/min = anomaly).
- **DS3231 Counterfeit Risk.** Genuine DS3231 achieves ±2 ppm (~63 ms/day drift). Counterfeits drift 50–100× faster. Verify authenticity before deployment.
- **DS3231 Polling Window Drift.** At ±2 ppm over 24 h: cumulative drift ~±170 ms — within the 2 s window. Hourly Master beacon re-synchronisation pulse recommended for long-term reliability.
- **BPW34 Radiation Detector.** Valid research technique but not a calibrated instrument. Not suitable for medical or regulatory compliance measurements. Suitable for environmental anomaly detection only.

Key References & Standards

- IETF RFC 3561 — *Ad hoc On-Demand Distance Vector (AODV) Routing*. C. Perkins et al., 2003.
- IETF RFC 3610 — *Counter with CBC-MAC (CCM)*. D. Whiting et al., 2003.
- NIST SP 800-38C — *Recommendation for Block Cipher Modes: CCM Mode*. NIST, 2004.
- LoRa Alliance — *LoRaWAN Regional Parameters v1.0.3*, IN865-867 Band Plan.
- Semtech — *SX1262 Datasheet Rev 2.1*; *LoRa Modulation Basics AN1200.22*.
- Espressif — *ESP32-S3 Technical Reference Manual v1.5*; *ESP32-S3 Datasheet v1.9*.
- Torex Semiconductor — *XC6220 Series Datasheet*.

- Consonance Semiconductor — *CN3722 Datasheet*.
- Matheron, G. (1963). *Principles of Geostatistics*. Economic Geology, 58(8), 1246–1266.
- Chen & Guestrin (2016). *XGBoost: A Scalable Tree Boosting System*. KDD 2016.